

ADS-B Assessment

September 2025



Serving a world in motion
navcanada.ca



Disclaimer

The contents of the attached presentation entitled “ADS-B Assessment – September, 2025” are proprietary and confidential to NAV CANADA and are being provided to Aircraft Electronics Association for internal purposes only.

This presentation shall not be circulated outside of Aircraft Electronics Association and is being provided to you on a confidential basis

Purpose

Provide an overview on ADS-B in Canada:

- Using the PAPR and DAPR
- Background on:
 - PU min (different than TCCA's PU min)
 - Long gaps figure
 - Comparison analysis and details
- Contact information to discuss issues with the tests
- Ground Based ADS-B
- Plans/Process for Class C/D/E airspace – estimated timeline

Purpose

Provide an overview of ADS-B in Canada

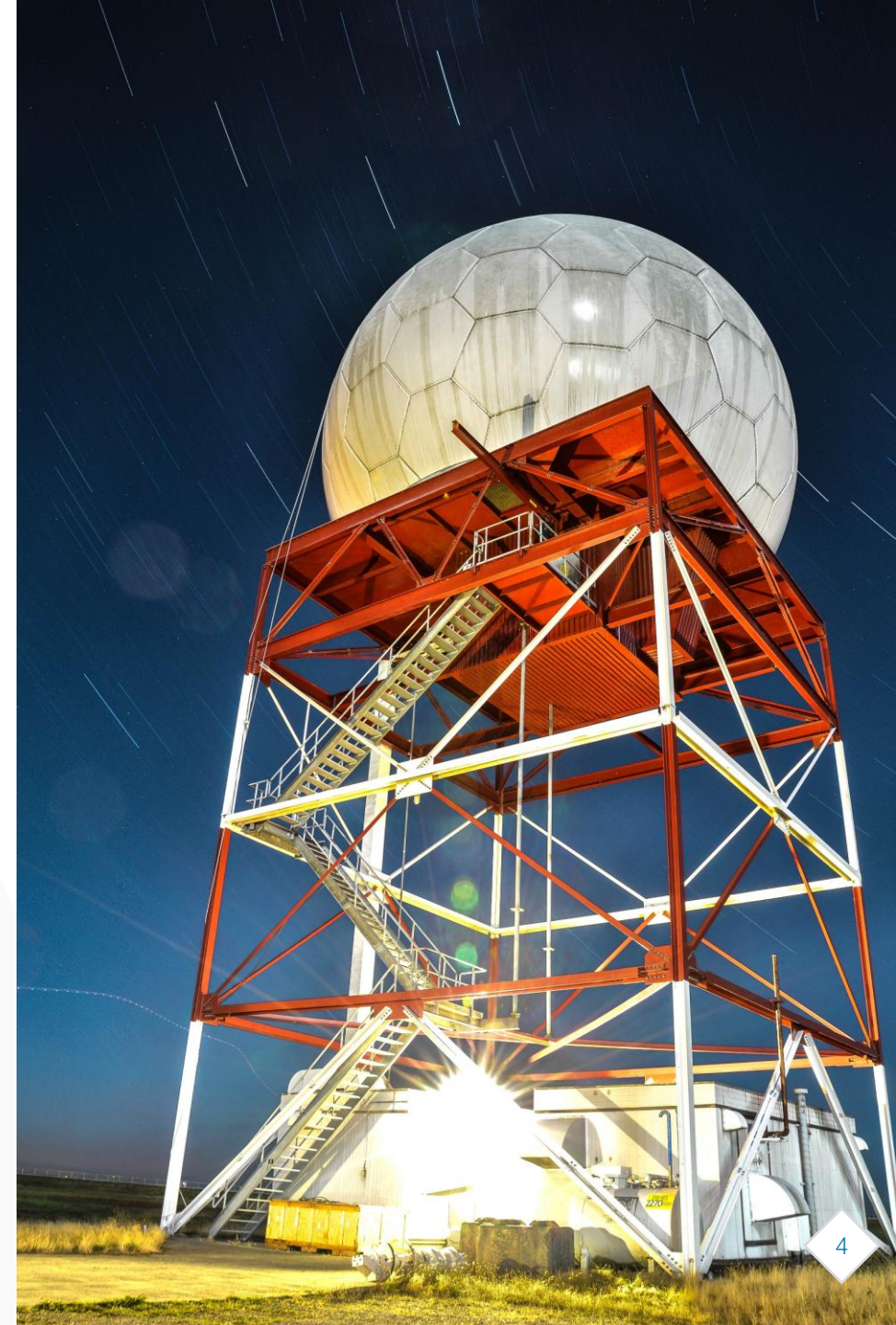
Using the PAPR and DAPR

- Background on:
 - PU min (different than TCCA's PU min)
 - long gaps figure
 - comparison Analysis and details
- Contact information to discuss issues with the tests

Ground Based ADS-B

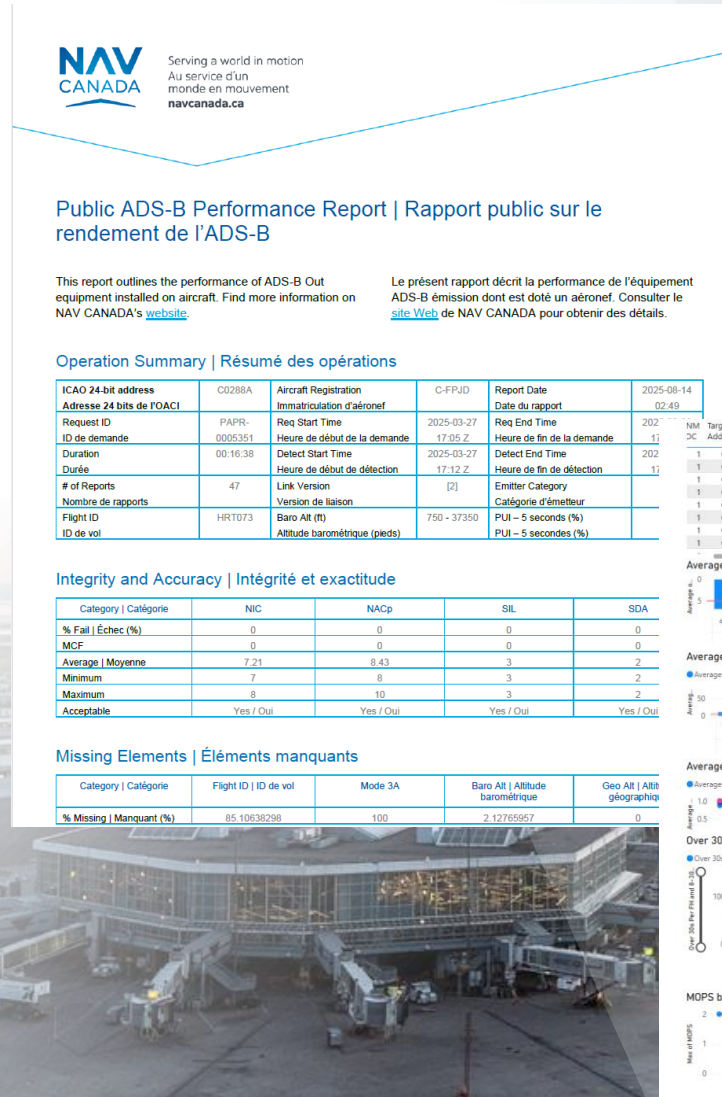
Plans/Process for Class C/D/E airspace

- estimated timeline

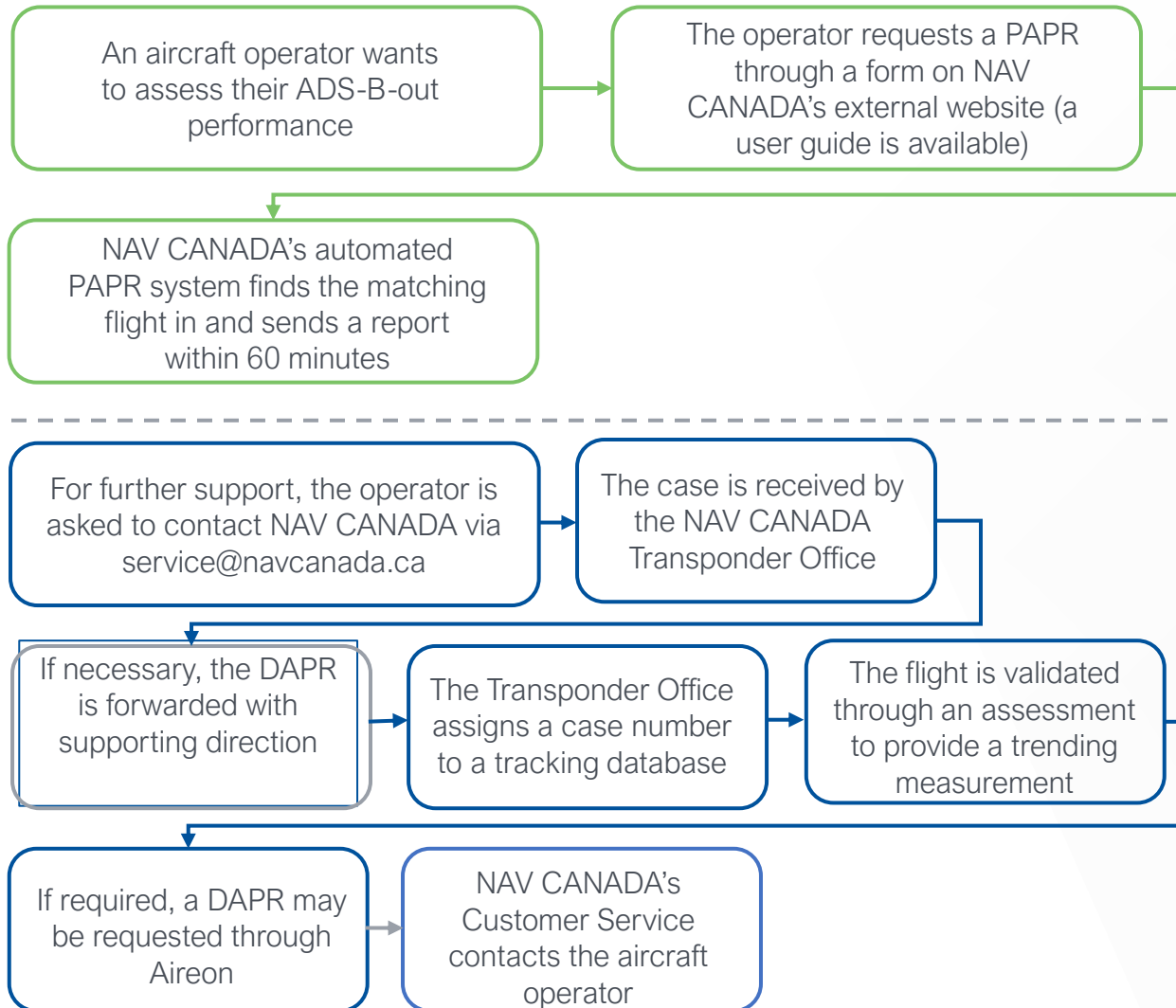


PAPR/Power BI/DAPR

- PAPR
- Power BI
- DAPR



The Public ADS-B Performance (PAPR) Report



PAPR Request Form

The screenshot shows the 'Public ADS-B Performance Report (PAPR) User Guide' document. It includes a NAV CANADA logo, a background section explaining the purpose of the report, and a version number of 1.0 - December 2023.

PAPR User Guide

The screenshot shows the 'PAPR User Guide' form. It includes sections for 'Flight Information' (ICAO Aircraft Address, Flight Start Date, Flight End Date) and 'Contact Information' (Contact Email Address, Email Address Confirmation). There is also a CAPTCHA section with a 'Generate a new image' button and a 'Submit' button.

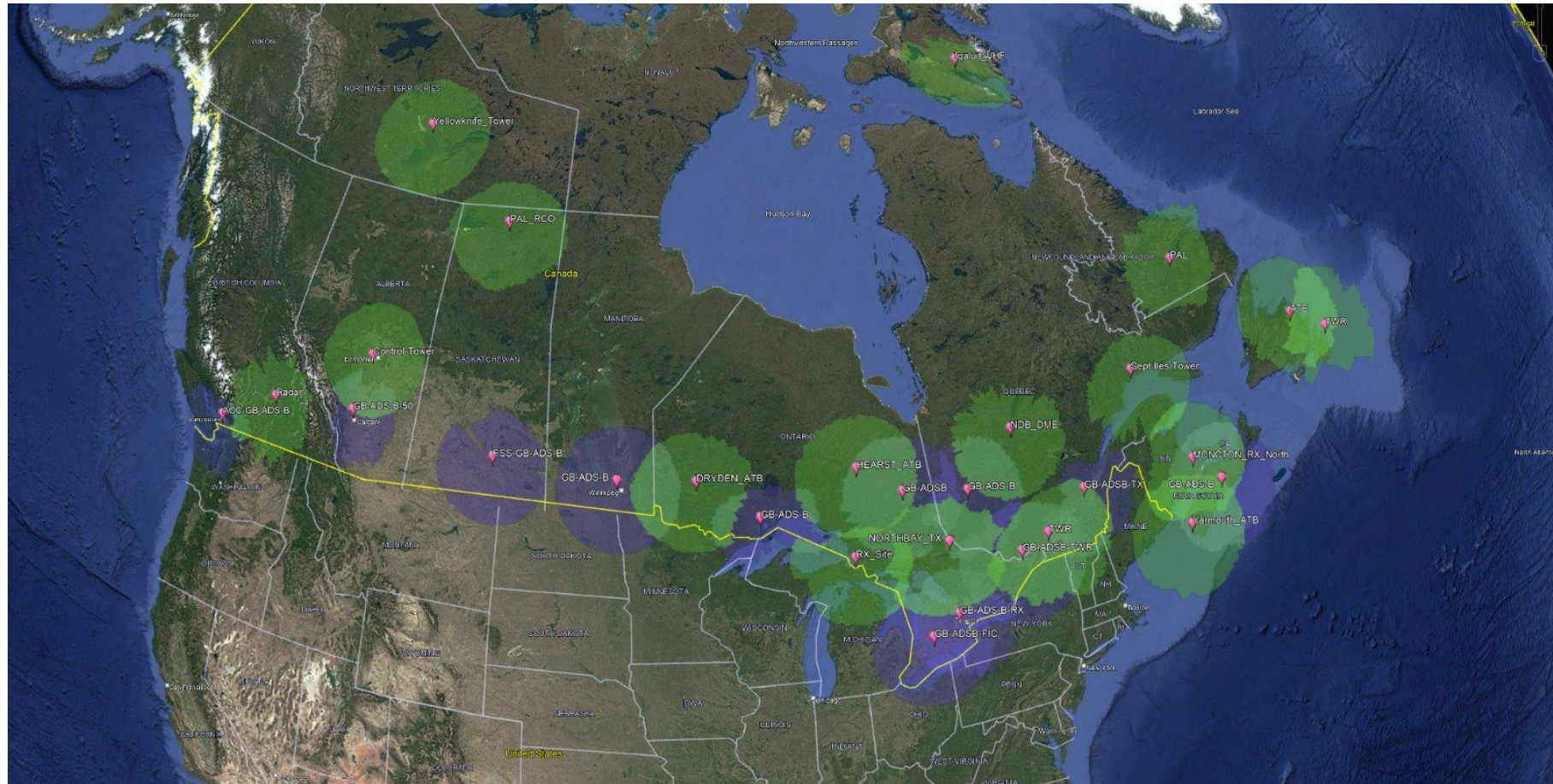
The DAPR is prepared by:

- Collecting all aircraft that have flown within 60nm at same time-frame of the test aircraft
- Using only aircraft that have historical PUI > 96.5% (ie, create a 'good performer list') – generate a PU Min. and create a baseline PU that is more resilient to interference
- Testing aircraft PUI displayed vs PU Min
- Detection characteristics (Asterix plots vs satellite elevation angle plot) helps identify if detection is mainly due to top or bottom antenna

Why FOM Matters

- Introduced in RTCA DO-260 MOPS (Figure of Merit)
 - Horizontal/Vertical FOM for position,
 - Horizontal/Vertical FOM for velocity,
 - along with NUCp/NUCv (position and velocity uncertainties)
 - NUC should have been based on integrity and not on accuracy in DO-260 (DO-260A statement)
- Replaced by a set of quality indicators in RTCA DO-260A MOPS and in later editions of the 1090 ADS-B out MOPS:
 - NAcP (Navigation Accuracy Category for position; dynamic value)
 - NAcv (Navigation Accuracy Category for velocity; dynamic value)
 - Along with
 - NIC (Navigation Integrity Code; dynamic value),
 - SIL (Surveillance Integrity Level; dynamic value)
 - SDA (Software Design Assurance; static value)
- NACp/SIL/NIC/SDA give an overall indication of the ADS-B data quality.
- PAPER reports provides an "Integrity and Accuracy" table which is made of NIC/NACp/SIL/SDA values
- Failing to meet quality indicators thresholds set out in AC 500-029 lead to filter out the ADS-B report from our ATM system

GB ADS-B Deployments – FL125



Plans/Process for C/D/E airspace

- In 2024 NAV CANADA announced that a Class C/D/E mandate would not be implemented before 2028
- NAV CANADA follows a coordinated communications process when launching studies to ensure broad awareness and accurate information sharing. As such, NAV CANADA cannot offer significant details on the proposed approach just yet.
- We encourage you to participate in this important consultation.



Questions ?

Serving a world in motion
navcanada.ca

